AutoSRE: an autonomous on-call engineer that proves its fixes

Team **Zero-Ops** · Problem statement: *Autonomous Incident Remediation: Closed-Loop Telemetry Diagnosis, Sandboxed Code Patching and GitOps Self-Healing*

The problem

When application code leaks database connections (for example, unclosed cursors or pooled connections that are never returned), PostgreSQL runs out of connections and the service goes down. Today an engineer is paged, manually correlates dashboards, logs and git history, writes a patch under pressure and documents it later. Typical recovery takes around 45 minutes. AI assistants only *suggest* code; they do not verify it, and nobody keeps their pull request mergeable while main keeps moving.

What AutoSRE does: a closed loop from alert to a mergeable, verified pull request

- **Detect:** a monitor (or any alert webhook) raises the incident when connections pass a threshold; duplicate alerts are suppressed.
- **Stop the bleeding:** ends only the stuck sessions of the affected app, using a least-privilege database role (it can read activity stats and end sessions, never read data). Service recovers in seconds.
- **Diagnose and find the cause:** groups stuck queries, parses logs, then scores recent commits against the evidence (query text, failing route, stack frames) and confirms the exact lines with *git blame*, with a confidence score.
- **Fix with our own coding agent:** a tool-using agent (read, search, edit, run tests, finish) writes a regression test that must **fail on the old code**, makes a minimal fix and iterates on test feedback (self-healing, max 3 attempts).
- **Verify in a sealed sandbox:** Docker with no network, no host mounts, non-root, CPU/memory/time limits; red-then-green proof before anything is delivered.
- **Deliver and keep it mergeable:** opens a draft PR with an auto-generated post-mortem, then watches it: if a teammate's commit causes a merge conflict or CI fails, the agent repairs it, re-tests and pushes, until the PR is ready. **A human always presses Merge.**

What makes it different

Capability	Typical AI SRE tools	AutoSRE
Fix verification	Confidence scores, or re-run a scanner	Regression test fails on old code and passes on new, in an isolated sandbox
After the PR opens	Waits for a human	Resolves merge conflicts and CI failures on its own PR until it is ready to merge
Safety rules	Policies around commands	Guardrails inside the agent's tools: file allowlist, 50-line cap, dangerous-code block, secret scan; refusals are fed back to the agent
AI security and ethics	Rarely described	Prompt-injection screening (heuristics + Llama Prompt Guard 2), secret and personal-data redaction before any LLM call, human review mode, Responsible-AI card and full audit trail per incident
Resilience	Needs the cloud model online	Model fallback chain, token pacing for free tiers, cached and replay modes clearly labelled

Demonstrated result

In the demo, a Django service leaks pooled PostgreSQL connections (98 of 100 used). AutoSRE ends 89 stuck sessions, pins the leak to commit `95f395a` ("perf(reports): add fast raw-SQL user activity report"), fails its first fix attempt, self-corrects to a context manager, verifies it, opens the PR, repairs a real merge conflict with a teammate's commit while keeping both changes, and the merged fix is pushed to GitHub. **Alert to verified, ready-to-merge fix in about 11 seconds** (cached mode) versus about 45 minutes manually.

Architecture and technology

Python 3.11, **Django 5 + DRF** (ASGI, Server-Sent Events), **LangGraph** state machine, our own tool-using coding agent on OpenAI-compatible LLMs (**NVIDIA Nemotron**, Groq gpt-oss, with fallback), PostgreSQL 16, GitPython, docker-py sandbox, PyGithub + **GitHub App** (bot identity, 1-hour tokens), React + TypeScript + shadcn/ui dashboard (light/dark, mobile). **Stack-independent:** any project describes how to build and test itself in `.autosre.yml` (or is auto-detected: Python, Node, Go, Java); databases plug in through adapters.

Links

Code: github.com/akshitsharmacodes/zero-ops-autosre · Demo target with the AutoSRE fix merged: github.com/akshitsharmacodes/victim-service